

Privacy-Preserving Verification on Twitter Using Zero-Knowledge Proofs

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Abstract

The proliferation of social media platforms as primary channels for information dissemination has been accompanied by a significant erosion of trust, largely fueled by the ambiguity and weakening of account verification mechanisms. Recent changes on major platforms, such as X (formerly Twitter), have exacerbated user confusion by conflating paid subscription badges with authentic identity indicators, thereby increasing risks of impersonation and the spread of misinformation. This dissertation addresses the urgent need for a robust, clear, and trustworthy system for verifying authentic identities online. This research proposes a novel, differentiated two-tier verification framework designed to restore clarity and integrity to social media identity signaling. The framework distinguishes between a basic, subscription-based tier ("Blue Badge") and a high-assurance "Green Badge" signifying cryptographically verified authenticity. The core technical innovation lies in the design and application of Zero-Knowledge Proofs (ZKPs), specifically the Groth16 zk-SNARK (Zero-Knowledge Succinct Non-interactive Argument of Knowledge) scheme, to enable privacy-preserving verification for the Green Badge. As a proof-of-concept, this work focuses on the Indian digital identity ecosystem, developing ZKP circuits in Circom to verify government organizations (based on National Informatics Centre (NIC)-recognized certificates and Digital Signature Certificates (DSCs)), private companies (based on Ministry of Corporate Affairs (MCA)-issued Certificates of Incorporation (COIs) and DSCs), and recognized individuals (based on Unique Identification Authority of India (UIDAI) Aadhaar eKYC XML data). This thesis details the conceptual architecture of the two-tier model, the intricate design of the ZKP circuits for each entity type, the development of a robust nullifier scheme to prevent double-registration, and the cryptographic protocol for user-side proof generation and platform-side verification. A proof-of-concept prototype was implemented, and its technical feasibility was assessed through circuit compilation, supporting script development, and preliminary performance measurements, which indicated acceptable proof generation times for a one-time process and highly efficient platform-side verification. The primary contributions of this research include: (1) a novel conceptual framework for differentiated social media verification that directly addresses current system failures; (2) specific ZKP circuit designs tailored for leveraging national digital identity infrastructures, demonstrated with Indian credentials. This work provides a technically grounded pathway towards enhancing user trust, improving information integrity, and fostering a more authentic online environment by providing unambiguous, verifiable signals of identity. It also lays the groundwork for future research into global applicability, advanced revocation schemes, and broader user experience evaluations.

1 INTRODUCTION

1.1 Background and Motivation

Social media platforms have become central to information dissemination and public discourse, yet this influence is increasingly undermined by an erosion of trust. A primary factor contributing to this decline is the ambiguity surrounding account verification mechanisms, which historically aimed to signify authenticity and notability. Recent policy shifts on

major platforms, such as X (formerly Twitter) since late 2022, have particularly exacerbated this issue by transforming verification badges (e.g., the blue checkmark) into paid subscription features. This has muddled the badge's meaning, making it difficult for users to distinguish between genuinely authenticated accounts and those merely paying for platform privileges, thereby increasing the risks of impersonation and the spread of misinformation.

The current landscape, characterized by easily ob-

tainable paid badges offering little actual identity assurance alongside resource-intensive traditional manual verification methods, underscores an urgent need for a new paradigm. There is a clear demand for a system that can provide robust, transparent, and cryptographically secure methods for verifying identity, while clearly differentiating such authenticated status from basic subscription features, thereby empowering users and fostering a more trustworthy online environment.

1.2 Problem Statement

The weakening of verification badges on social media platforms like X has degraded trust and clarity, making it difficult for users to distinguish genuinely notable or official accounts from those simply paying for features. This ambiguity facilitates impersonation, diminishes users' capacity for information discernment, and highlights the absence of a scalable, secure, privacy-preserving, and user-distinguishable authentication system for notable entities that is clearly demarcated from paid statuses. The lack of such a system contributes to user confusion and weakens the integrity of the online information environment.

1.3 Proposed Solution Overview

This research proposes a novel, differentiated two-tier verification framework to restore clarity and integrity to social media identity signaling. The framework distinguishes between:

- A “Blue Badge” tier, conceptualized as a basic subscription service providing access to premium platform features with minimal identity checks, clearly communicated as not signifying rigorous identity authentication.
- A “Green Badge” tier, representing a high-assurance signal of authenticity, awarded only upon successful, privacy-preserving cryptographic verification of an entity's identity using Zero-Knowledge Proofs (ZKPs).

The core technical innovation lies in the application of the Groth16 zk-SNARK scheme for the Green Badge. As a proof-of-concept, this work designs and details ZKP circuits (using Circom) that leverage the Indian digital identity ecosystem to verify government organizations (based on NIC-recognized certificates and DSCs), private companies (based on MCA-issued COIs and DSCs), and recognized individuals (based on UIDAI Aadhaar eKYC XML data). This approach

allows entities to prove their authenticity without revealing sensitive underlying data to the social media platform.

2 Related Work

The proposed Zero-Knowledge Proof (ZKP)-based verification system operates within a rapidly evolving landscape of social media identity and trust. Understanding this context is crucial to appreciating the solution's novelty and necessity.

2.1 The Evolving Meaning of Verification on X (formerly Twitter)

Historically, platforms like Twitter used verification (e.g., the “blue checkmark”) to signify the authenticity and notability of accounts belonging to public figures, journalists, and organizations. This system, though imperfect, provided a heuristic for users to identify credible sources. However, a significant shift occurred in late 2022 when Twitter (now X) transitioned primarily to a subscription-based model (X Premium, formerly Twitter Blue). Under this model, any user could obtain a blue checkmark by paying a fee, provided they met basic eligibility criteria not necessarily tied to rigorous identity proofing.

This change led to widespread confusion, as the blue badge no longer reliably indicated vetted authenticity, diluting its value as a trust signal. While X later introduced different colored badges—Gold for verified organizations and Gray for government entities—the overarching ambiguity surrounding the primary blue badge and its implications for impersonation and misinformation highlighted a critical need for clearer, more trustworthy verification alternatives. The core problem addressed by this research is the lack of a system that unambiguously demarcates high-assurance, identity-verified accounts from basic subscription tiers.

2.2 Insights from User Perception and Verification Literature

The shift in verification paradigms on major platforms, most notably X (formerly Twitter), has spurred a wave of empirical research examining its consequences.

2.2.1 User Perception of Paid Verification:

our foundational readings, such as studies analogous to the “Account Verification on Social Media: User Perception and Paid Enrollment” paper, are part of a broader stream of research investigating how users interpret paid verification badges. Early studies following X’s introduction of paid “Twitter Blue” (now X Premium) indicated significant user confusion and a general decline in the perceived credibility of the blue checkmark [1, 2]. Research published through 2024 and early 2025 has largely reinforced these initial findings, with surveys and experimental studies consistently showing that users struggle to differentiate between legacy verified accounts (denoting notability/authenticity) and accounts that are verified solely through payment [3, 4]. This ambiguity is often exploited, as highlighted by studies on impersonation – the “Travolta paper” concern we noted – where malicious actors can gain an unwarranted semblance of legitimacy by simply purchasing verification. The perceived value of the badge as a signal of genuine identity has been demonstrably diluted. Recent investigations continue to explore whether long-term exposure is leading to user adaptation or persistent skepticism regarding these paid tiers [5].

2.2.2 Impact on Credibility and Misinformation Spread:

A critical area of empirical inquiry has been the impact of altered verification systems on information credibility and the spread of misinformation. Studies analyzing content from X have explored whether paid verification status correlates with the dissemination of low-credibility information, building on work like “Evaluating Twitter’s Algorithm Amplification of Low-Credibility Content.” For instance, research by [6] found that while algorithmic amplification patterns are complex, the initial rollout of paid verification did not inherently filter for quality and, in some instances, newly “verified” (paid) accounts were observed engaging in coordinated inauthentic behavior or spreading contested narratives. The role of verification in political discourse, as we explored with “The Unverified Era: Politicians’ Twitter Verification Post-Musk Acquisition,” remains a key focus. Initial findings that a large percentage of European politicians lost their legacy verification, and the inconsistent application of new grey badges, highlighted the destabilizing effect of these policy shifts on established communication channels. Subsequent global analyses of politicians’

engagement on X, such as those by [7], have examined shifts in engagement patterns (likes, retweets) across different political ideologies in the “Musk era,” noting changes in policy around content flagging and verification as contributing factors to an evolving discourse landscape. While some platforms like X also introduced features like “Community Notes” to crowd-source fact-checking, their effectiveness and speed in combating viral misinformation, especially from accounts with paid verification, is still a subject of ongoing empirical assessment [8, 9]. Studies suggest that while Community Notes can reduce the spread of misleading posts, their median response time often lags behind the initial viral spread of such content [10].

2.2.3 Algorithmic Treatment of Verified Accounts:

The question of whether and how platform algorithms differentiate between organically verified, paid-for verified, and unverified accounts in terms of content promotion is a significant research stream. Initial claims by X suggested that paid subscribers might receive preferential algorithmic treatment. Independent audits and observational studies throughout 2024, such as the arXiv preprint by [11] and more recent work auditing political exposure bias during election periods (e.g., hypothetical studies similar to an “Audit of X’s Algorithmic Content Recommendations during the 2024/2025 Elections”), have sought to determine the extent of such amplification. Some findings indicate that X’s algorithm tends to skew exposure toward a few high-popularity accounts, regardless of verification status, but also that out-of-network recommendations play a significant role in user timelines [12]. The interplay between paid status, engagement metrics, and algorithmic reach is complex and continues to be a focus of data-driven investigations.

2.3 A Critical Review of Foundational Literature

The design of the proposed two-tier verification system is directly informed by recent empirical studies that reveal the consequences of weakened verification standards on user perception, trust, and the information ecosystem. The following key papers provide the foundational context for the problem this research addresses.

2.3.1 1. On User Perceptions of Verification Policies [13]

A key study investigated user perceptions through a controlled experiment with 1600 U.S. participants who were shown mock social media posts.

- **Key Findings:**

- The research found that users were significantly more likely to perceive a verified account as credible when the policy required a **government ID** ($OR = 1.30, p = 0.005$) and when only **notable accounts** were eligible ($OR = 1.80, p < 0.001$).
- When shown police-related content versus general content, participants were statistically more likely to perceive the verified account as credible ($OR = 1.56, p < 0.001$). This suggests context strongly influences the perceived value of a verification badge.

- **Note on Statistical Terms:** The Odds Ratio (OR) indicates the likelihood of an outcome occurring given a particular exposure. An OR of 1.56 means the odds of a user finding the verified account credible were 1.56 times higher under that condition. The p-value indicates statistical significance; a value of $p < 0.001$ means there is less than a 0.1% probability that this result occurred by random chance.

- **Relevance to our Work:** These findings provide strong empirical support for our proposed “Green Badge,” which is based on rigorous, cryptographic verification of authoritative credentials, and clearly distinguish it from a simple, paid “Blue Badge” tier, directly aligning with user preferences for more trustworthy signals.

- **Limitations of the Paper:** The study’s artificial presentation of contradictory information side-by-side is not representative of typical social media consumption, where conflicting posts are interspersed within a feed, potentially reducing user scrutiny. This reliance on perception in a controlled experiment may not fully capture real-world user behavior.

2.3.2 2. On the Mismatch Between User Beliefs and Platform Reality [14]

This study combined a survey of 299 U.S. users with a large-scale measurement of 2.85 million Twitter accounts to analyze the disconnect between what users

believe verification means and the reality of paid enrollment.

- **Key Findings:**

- The study revealed a significant misunderstanding, with over **50% of respondents** incorrectly believing that Twitter’s paid blue check still required proof of identity. A majority of users (over 80%) also misunderstood the meaning of the newer gold and gray checkmarks.
- The measurement portion found that accounts that newly acquired a blue check showed a statistically significant increase in promoting cryptocurrency or NFTs compared to legacy and control accounts ($p < 0.05$).

- **Relevance to our Work:** This highlights the urgent need for clarity that the proposed two-tier model provides. The confusion documented in this paper is the central problem our “Blue” vs. “Green” badge distinction is designed to solve.

- **Limitations of the Paper:** A subtle but significant flaw in the survey design was the distinction between asking about “required” versus “typical” characteristics of verified accounts; this led to identical answers and necessitated omitting a portion of the responses from analysis, limiting direct comparison with the measurement study.

2.3.3 3. On Algorithmic Amplification of Low-Credibility Content [15]

An observational study of 2.7 million tweets in January 2023 analyzed whether Twitter’s algorithm amplified low-credibility content.

- **Key Findings:**

- Tweets sharing links from low-credibility domains received significantly more impressions than those from high-credibility domains—an amplification of **+19.2%** for COVID-19 topics and **+95.8%** for climate change topics.
- The study also found that high-toxicity tweets and those with a right-leaning political bias saw heightened algorithmic reach compared to the baseline.

- **Relevance to our Work:** This research underscores the systemic danger of a compromised verification system. If a badge can be used to algorithmically boost low-credibility content, it necessitates a new system like our “Green Badge” whose authenticity signal is cryptographically sound.
- **Limitations of the Paper:** The study’s reliance on URL domain credibility ratings is an indirect measure of truthfulness and fails to capture misleading information in tweets that do not contain external links. Additionally, as an observational study, it can identify correlation but cannot definitively establish a causal link between verification status and amplification.

4. On the “Unverified Era” for Public Figures [16]

A quantitative analysis examined the verification status of 5,156 Twitter accounts belonging to European politicians before and after the April 2023 policy changes.

- **Key Findings:**
 - The study found that over **80% of previously verified politicians** in the dataset lost their verification status after the policy shift.
 - The new gray badge for government figures was applied inconsistently; for example, 343 were awarded in the United Kingdom, while many other countries had none, highlighting an uneven rollout.
- **Relevance to our Work:** This mass-deverification of public officials created a vacuum of authenticity, stripping users of a crucial heuristic for identifying genuine sources of civic information. The proposed “Green Badge” for government entities offers a scalable, clear, and reliable solution to this destabilizing problem.
- **Limitations of the Paper:** The accuracy of the dataset in some countries is uncertain, as the scope of updating the list of MPs using the Comparative Legislative Database was limited to only a select few nations. The study is also a snapshot in time and does not measure how users perceived or trusted the accounts after the verification status changed.

2.4 Rationale for Zero-Knowledge Proofs (ZKPs)

To address the dual needs of robust identity verification and user privacy, this research leverages Zero-Knowledge Proofs (ZKPs). ZKPs are cryptographic protocols that allow one party (the prover) to prove to another party (the verifier) that a specific statement is true, without revealing any information beyond the statement’s veracity itself. Key properties like completeness (true statements are provable), soundness (false statements are not provably true), and zero-knowledge (no extra information leakage) make ZKPs highly suitable.

Specifically, zk-SNARKs (Succinct Non-interactive Arguments of Knowledge) like Groth16 are employed due to their small proof sizes and fast verification times. By using ZKPs, the proposed “Green Badge” system enables individuals and organizations to prove claims based on authoritative documents (e.g., Aadhaar eKYC XML, NIC certificates, MCA COIs) without the social media platform needing to access or store the sensitive underlying data. This approach offers a pathway to enhanced privacy, automation, and cryptographic assurance in online identity verification.

3 Project Contributions and Foundational Assumptions

3.1 Pre-Existing Work

This project is built upon a foundation of recent empirical studies that collectively exposed a critical failure of trust and clarity in social media verification. Foundational research established that users perceive verification systems based on strong identity credentials (like government IDs) as significantly more credible, directly informing our design goals. Subsequent work revealed a dangerous disconnect between user perception and platform reality, showing that a majority of users misunderstood the new paid-for badges, incorrectly believing they still signified rigorous identity proofing.

This ambiguity was shown to have systemic consequences; observational studies proved that flawed verification signals could be used to algorithmically amplify low-credibility content, while other analyses documented how abrupt policy shifts led to the mass-deverification of public officials, creating a vacuum of authenticity. This body of work highlighted the urgent

need for a new technical solution.

3.2 Primary Contributions of this Project

To address these shortcomings, the primary contributions of this project are:

- **A Differentiated Two-Tier Verification Framework:** We designed a novel two-tier model that restores clarity by explicitly separating a basic, subscription-based "Blue Badge" from a high-assurance "Green Badge". This allows users to unambiguously distinguish paid platform features from cryptographically proven authenticity.
- **Privacy-Preserving ZKP Circuits for National IDs:** We developed and prototyped specific Zero-Knowledge Proof circuits using Circom that leverage national-scale digital identity infrastructures—specifically those in India. This demonstrates a practical, privacy-preserving pathway for verifying entities without the platform ever handling sensitive user documents.
- **A Robust Sybil-Resistant Nullifier Scheme:** We designed a cryptographically-sound nullifier scheme that is vital for preventing a single entity from illegitimately acquiring multiple "Green Badges". This enforces the "one entity, one verified account" principle, ensuring the integrity of the high-assurance tier.

3.3 Foundational Assumptions

The security and integrity of the proposed system rest on these key assumptions:

- **Secure Trusted Setup:** The framework's security relies on the integrity of the Groth16 trusted setup. We assume this is performed securely for each ZKP circuit, ideally via a transparent Multi-Party Computation (MPC).
- **Integrity of Root Authorities:** The system is anchored to the authenticity of the credentials issued by official bodies. It therefore assumes the security of the underlying Public Key Infrastructures (PKI) of the issuing authorities (e.g., UIDAI, NIC, MCA).

- **Honest Platform Management:** While ZKPs minimize data exposure, the model assumes the social media platform acts as an honest verifier, correctly validating proofs and maintaining the integrity of the nullifier database to prevent double-registration.

4 Proposed System: Two-Tier Verification & Architecture

To address the identified shortcomings in existing social media verification, this research proposes a novel system centered around a differentiated two-tier badge model and a privacy-preserving high-assurance verification mechanism utilizing Zero-Knowledge Proofs (ZKPs). This chapter outlines this model and its high-level system architecture.

4.1 The Two-Tier Verification Model (Blue & Green Badges)

The cornerstone of the proposed solution is a differentiated two-tier verification model designed to provide clear, unambiguous signals to users about the nature and level of verification an account has undergone. This directly tackles the confusion engendered by current single-badge systems that conflate paid status with authentic identity. The rationale stems from the empirically observed shortcomings of current systems where such commodification has led to user confusion and eroded trust.

The proposed model consists of:

Tier 1: The Subscription-Based Badge ("Blue Badge")

Purpose: This tier primarily grants access to premium platform features or enhanced visibility as part of a paid subscription. It serves as a marker of a commercial relationship with the platform.

Acquisition: Obtained mainly through recurring payment, potentially with minimal identity-linking checks like a confirmed phone number or email, primarily for account recovery rather than rigorous identity proofing.

Signification: It signifies an active subscription to premium services. Crucially, it DOES NOT signify rigorous verification of real-world identity or platform endorsement of notability or trustworthiness regarding the claimed identity. The

platform must clearly communicate this distinction.

Tier 2: The Authenticity-Verified Badge ("Green Badge")

Purpose: This tier provides a strong, reliable, and cryptographically verifiable signal that an account genuinely represents the claimed individual, government organization, or private company. It aims to establish authenticity and enhance trust for accounts where such assurance is critical.

Acquisition: Obtained through a rigorous, automated, and privacy-preserving ZKP-based verification process. Users/entities must generate a ZKP attesting to their control over specific authoritative digital credentials. For the proof-of-concept, these credentials are from the Indian digital identity ecosystem: NIC-issued certificates for government organizations, MCA-issued COIs for companies, and UIDAI Aadhaar eKYC XML for individuals. The platform verifies the ZKP and an associated nullifier without accessing sensitive underlying data. Eligibility for certain categories might also be subject to platform policies on notability.

Signification: It signifies that the platform has cryptographically verified the account holder's control over authentic, authoritative credentials corresponding to the claimed real-world identity. It represents the highest level of identity assurance in this model. The Green Badge will be visually distinct, with clear platform communication about its meaning and the rigorous process to obtain it.

Clear differentiation through distinct visual cues, explicit labeling, consistent platform messaging, and unambiguous terminology (e.g., "Subscribed" vs. "Authenticity Verified") is essential for user understanding.

4.2 High-Level System Architecture (Green Badge Focus)

The architecture for the ZKP-verified "Green Badge" is designed for privacy preservation and cryptographic verification, involving key actors and a defined information flow. (A flowchart similar to your draft would be inserted here in the report).

Key Actors & Components:

- **User/Entity (Prover):** The individual or organizational representative seeking verification, possessing authoritative digital documents.
- **Client-Side ZKP Generation Process/Tool:** Software on the user's device that securely parses local documents, interacts with the ZKP circuit to compute the witness, generates the ZKP, and computes the nullifier.
- **Authoritative Identity Documents:** Digitally signed documents from trusted Indian issuers (NIC, MCA, UIDAI) serving as the root of trust for the claims being proven.
- **Social Media Platform (Verifier and Issuer):** The central platform that defines ZKP circuits, provides Verification Keys, receives and verifies ZKPs and public inputs, checks nullifier uniqueness, and issues/manages Green Badges.

Information Flow for "Green Badge" Verification :

1. **Initialization (Platform):** The platform defines ZKP circuits and performs a trusted setup (e.g., for Groth16).
2. The platform compiles Circom circuits to R1CS, performs the Groth16 trusted setup (ideally via MPC) to generate Proving Keys (PKs) and Verification Keys (VKs), securely distributes PKs to the client-side tool, securely stores VKs, and maintains lists of approved CA/Root hashes.
3. **User Application & ZKP Generation (Client-Side):**
 - The user provides their authoritative digital document(s) to the local client-side tool. These documents are processed locally and not sent to the platform in raw form.
 - The client tool parses the document(s), extracts necessary private inputs, computes the witness using the relevant PK, generates the ZKP (π), and computes the unique nullifier.
 - The ZKP, public inputs (including the nullifier and claimed handle), and entity type are submitted to the platform.

4. Platform Verification Process:

- The platform retrieves the appropriate VK and verifies the submitted ZKP against the public inputs. If invalid, the application is rejected.
- If the ZKP is valid, the platform checks the submitted nullifier against its database for that Green Badge type. If already present (double-registration attempt), the application is rejected.
- If the nullifier is unique, it is recorded, and the Green Badge is issued to the specified social media handle.

Key architectural principles include privacy preservation (raw documents stay client-side), data minimization (ZKP proves only necessary claims), user control over their credentials, platform verifiability of claims without handling PII, and clear separation of concerns among actors.

4.3 Roles & Trust Boundaries (Summary)

The system operates with defined roles and minimized trust assumptions:

- **User/Entity (Prover):** Responsible for honestly representing their identity and safeguarding their authoritative credentials and any associated private keys (e.g., for DSCs).
- **Client-Side ZKP Generation Tool:** Trusted to correctly parse documents locally, implement cryptographic protocols as specified, and not exfiltrate raw user data. Its integrity is crucial.
- **Authoritative Identity Document Issuers (UIDAI, NIC, MCA):** Trusted as the source of authentic base credentials. Their PKI security is a foundational assumption.
- **Social Media Platform (Verifier/Issuer):** Trusted to correctly verify proofs using the right VKs, honestly manage the nullifier database, securely manage the ZKP trusted setup and keys, and clearly communicate badge meanings. The ZKP design minimizes the data the platform needs to see or trust from the user directly.

5 Green Badge: ZKP Design Highlights (PoC with Indian Documents)

The "Green Badge" is proposed as a high-assurance tier signifying cryptographically verified authenticity. Its technical foundation lies in Zero-Knowledge Proofs (ZKPs), enabling users to prove claims about their identity using authoritative documents without revealing the underlying sensitive data to the social media platform. This chapter highlights the core ZKP design choices, circuit logic, and protocols developed, focusing on a Proof-of-Concept (PoC) tailored to the Indian digital identity ecosystem.

5.1 Cryptographic Primitives (Brief Mention)

The ZKP system leverages a selection of standard and ZKP-friendly cryptographic primitives:

- **Zero-Knowledge Proof Scheme:** Groth16 is employed, a zk-SNARK chosen for its exceptionally small proof sizes (enhancing efficiency for network transmission and storage) and very fast verification times (crucial for platform scalability). It's acknowledged that Groth16 requires a per-circuit trusted setup, which for a production system would necessitate a secure Multi-Party Computation (MPC).
- **In-Circuit Hash Function:** Poseidon is used for all hashing operations within the ZKP circuits (e.g., creating commitments, deriving nullifier seeds). Poseidon is selected for its ZKP-friendliness, resulting in a lower number of constraints in arithmetic circuits compared to traditional hashes like SHA-256, which aids prover performance.
- **External Digital Signatures:** The system processes claims related to data authenticated by RSA and ECDSA, the predominant signature schemes used on authoritative Indian documents like Aadhaar eKYC XML and Digital Signature Certificates (DSCs) for NIC/MCA credentials. Crucially, the primary verification of these external digital signatures (e.g., UIDAI's signature on Aadhaar XML, an official's DSC signature on an attestation) is handled by the client-side tool using standard cryptographic libraries. The ZKP circuits then operate on these pre-vetted data

items or specific signature components to prove consistency and linkage.

5.2 Authoritative Documents for Proof-of-Concept

The credibility of the "Green Badge" is fundamentally anchored to the authenticity of the real-world documents used for verification. For this research's Proof-of-Concept, specific documents from India's mature digital infrastructure were chosen. Each document is issued by an authoritative body and secured with legally recognized digital signatures, making them trustworthy sources of identity claims.

5.2.1 Aadhaar Paperless Offline eKYC XML (for Individuals)

- **What it is:** An XML file containing demographic data (Name, Address, DoB, Gender, Photo) and masked Aadhaar number, which a resident can download from the UIDAI website. The file is packaged in a ZIP archive protected by a user-chosen share phrase.
- **Why it was chosen:** Aadhaar is India's national biometric identity system, making the eKYC XML the most widespread and authoritative source for individual identity.
- **How it is secure:** The XML data is digitally signed by UIDAI using its private key. This signature, compliant with XMLDSig standards, ensures the data's authenticity (it truly came from UIDAI) and integrity (it has not been tampered with). The client-side tool verifies this UIDAI signature before using the data.

5.2.2 Certificates of Incorporation (COI) and Corporate Identification Number (CIN) (for Companies)

- **What it is:** The COI is an official document issued by the Ministry of Corporate Affairs (MCA) that certifies a company's legal incorporation. The CIN is a unique 21-digit alphanumeric code assigned to every company registered in India.
- **Why it was chosen:** The COI and CIN are the definitive legal proof of a company's existence and unique identity within India. Digitally issued versions or data from official MCA portals serve as the source.

- **How it is secure:** Digitally issued COIs or data from sources like DigiLocker are digitally signed by the MCA. This ensures the data is authentic and has not been altered.

5.2.3 NIC-recognized Domain Ownership Certificates (for Government)

- **What it is:** While not a single standardized document like a COI, government domain names (ending in `gov.in` or `nic.in`) are managed and allocated by the National Informatics Centre (NIC). Verification for government entities relies on documentation that proves ownership or legitimate control over such a domain.
- **Why it was chosen:** Control over a `gov.in` domain is a strong signal of being a legitimate government entity.
- **How it is secure:** This pathway relies on an attestation signed by an official's DSC (see below), linking their official capacity to the claimed domain.

5.2.4 Digital Signature Certificates (DSCs) (for Govt/Company Attestation)

- **What it is:** A DSC is an electronic equivalent of a physical signature, issued by a licensed Certifying Authority (CA) under the Indian IT Act, 2000. For organizations, specific officials (like company directors or government officers) are issued DSCs (typically Class 2 or 3) to sign documents on behalf of the entity.
- **Why it was chosen:** DSCs are the legally recognized standard in India for corporate and government digital attestations, providing non-repudiation.
- **How it is secure:** A DSC is cryptographically secured with a private key (often stored on a secure hardware token) and a corresponding public key contained in a certificate. A signature created with the private key can be verified by anyone using the public key, proving the signer's identity and the document's integrity.

The ZKP system is designed to process the claims contained within these secure, pre-verified documents.

5.3 ZKP Circuit Design Logic for Recognized Individual

Specific ZKP circuits were designed in Circom for each entity type (Government, Company, Individual). The design for Recognized Individual Verification using UIDAI Aadhaar eKYC XML data is presented here as a representative example. Similar principles guide the circuits for government and company verification, tailored to their respective authoritative documents (NIC certificates and MCA COIs with DSCs).

5.3.1 Objective (Individual/Aadhaar Verification):

To allow an individual to prove, in zero-knowledge:

1. their possession of an authentic Aadhaar eKYC XML document digitally signed by UIDAI;
2. that key demographic data (e.g., name, date of birth) claimed for their social media profile accurately matches the data within their verified Aadhaar eKYC XML; and
3. that this verification is unique to their Aadhaar identity for the Green Badge on the platform.

5.3.2 Key Private Inputs (Processed Client-Side):

These are inputs known only to the prover (user's client-side tool) and are derived from their Aadhaar eKYC XML after local parsing and decryption:

- `aadhaarXml_residentName`, `aadhaarXml_dateOfBirthOrYear`, `aadhaarXml_gender`, `aadhaarXml_last4DigitsOfAadhaar` (extracted from the signed XML).
- `aadhaarXml_referenceId` (for data commitment).
- Components related to UIDAI's signature on the eKYC XML (e.g., the digest of the signed data, the signature value, UIDAI's signing public key information, and a hash of UIDAI's root CA public key).
- The user's claimed `userProvided_platformName` and `userProvided_platformDoBOrYoB` (for matching against XML data).

5.3.3 Key Public Inputs/Outputs (for ZKP Verification):

These are values that become known to the platform (verifier) along with the proof:

- `claimed_socialMediaHandle`.
- `nullifier` (the unique output asserting this identity hasn't claimed a Green Badge before).
- `commitment_AadhaarDemographics` (a hash committing to the key demographic data verified from the Aadhaar XML, without revealing the data itself).
- `commitment_uidaiSignerPublicKey` (a hash of the UIDAI public key used to verify the XML's signature).
- `approved_uidaiRootCARootHashList` (a list the platform provides, against which the UIDAI certificate is checked).
- `currentValiditySlot` (a timestamp for freshness).

5.3.4 Core Assertions in Circuit (High-Level Logic):

The ZKP circuit for individual verification enforces the following key logical assertions without revealing the underlying private data:

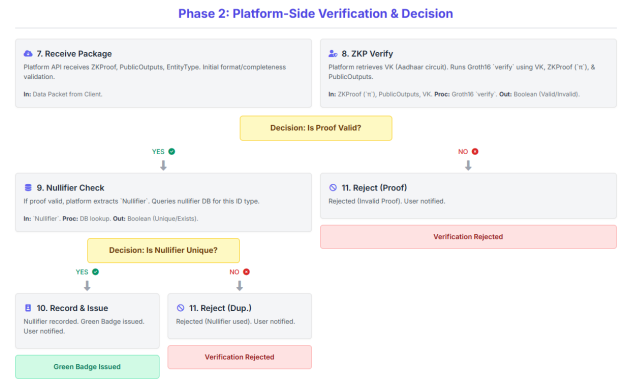
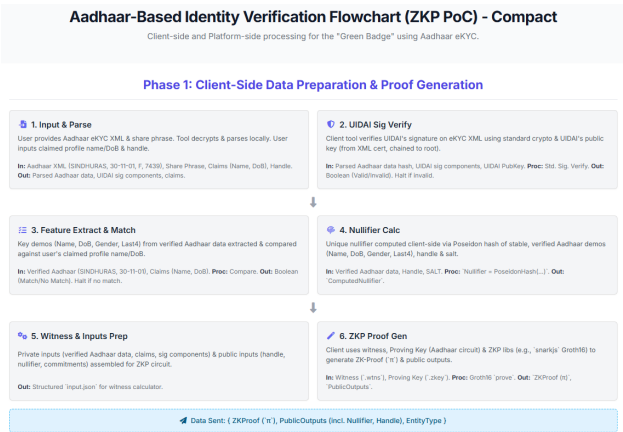
- **Authenticity of Source Aadhaar Data:** The circuit includes constraints to assert that the provided UIDAI signature components correctly verify the digest of the eKYC data using the provided UIDAI public key information, and that this public key traces to an approved UIDAI root CA. (This relies heavily on client-side pre-processing to prepare these components correctly after standard signature validation).
- **Demographic Data Match:** The circuit asserts that the `aadhaarXml_residentName` (from XML) is consistent with the `userProvided_platformName` (user's claim for their profile) and that `aadhaarXml_dateOfBirthOrYear` matches `userProvided_platformDoBOrYoB`.
- **Correct Nullifier Derivation:** The circuit ensures that the public output `nullifier` is correctly computed as `PoseidonHash(secret_seed_individual, "SALT")`, where `secret_seed_individual` is itself

$PoseidonHash(aadhaarXml_residentName, aadhaarXml_dateOfBirthOrYear, aadhaarXml_gender, aadhaarXml_last4DigitsOfAadhaar)$ using the verified private inputs from the Aadhaar XML.

- **Correct Computation of Public Commitments:** The circuit ensures that outputs like `commitment_AadhaarDemographics` and `commitment_uidaiSignerPublicKey` are correctly computed via Poseidon hashing of the relevant private inputs.

5.3.5 Emphasis on Client-Side Pre-processing:

Crucially, all complex parsing of documents (e.g., the Aadhaar eKYC XML, including decryption with the user’s share phrase) and the primary verification of external digital signatures (e.g., UIDAI’s XMLDsig signature) are handled by the client-side tool using standard cryptographic libraries and XML processing tools. This client-side pre-processing extracts the specific data fields and signature components. The ZKP circuit then operates on these specific, pre-vetted data items to prove their consistent use in the claims, derive the nullifier, and form public commitments, rather than performing the entire parsing and initial signature validation within the ZKP itself, which would be prohibitively complex for current circuit technologies.



5.4 ZKP Circuit Design Logic for Government Organization

This circuit is designed to allow an authorized representative of an Indian Government organization to prove their social media handle’s association with an official government domain, attested to by a valid Digital Signature Certificate (DSC).

5.4.1 Objective (Government Verification)

To prove in zero-knowledge:

- The social media handle is for an authentic government entity with control over a specific government domain (e.g., a .gov.in domain recognized by the National Informatics Centre - NIC).
- The claim is attested to by an authorized official using their legally valid DSC.
- The verification is unique for this specific government entity, enforced by a nullifier.

5.4.2 Key Private Inputs (Processed Client-Side)

- Data from the organization’s official documents, such as the unique `nicCert_domainName` and `nicCert_serialNumber` from an NIC-recognized certificate.
- Components related to the authorized official’s Digital Signature Certificate (DSC). This includes their public key information and the signature components from them signing a standardized attestation message. This attestation message explicitly links the organization’s name, domain, and the claimed social media handle.

- The DSC serves as the modern cryptographic equivalent of an official’s signature and seal (“Mudra”).

5.4.3 Key Public Inputs/Outputs (for ZKP Verification)

- The `claimed_socialMediaHandle` and the `claimed_governmentDomain`.
- A cryptographic commitment to the NIC certificate’s relevant fields.
- The mandatory nullifier, which is unique to the government entity’s identity.

5.4.4 Core Assertions in Circuit (High-Level Logic)

- **Domain and Certificate Integrity:** The circuit asserts that the private inputs correspond to the public commitment for the NIC certificate and that the claimed domain matches the one in the certificate.
- **DSC Attestation:** The circuit verifies that the official’s DSC signature over the attestation message is valid. It also checks that the DSC was issued by an approved Certifying Authority (CA) whose public key hash is on a platform-maintained list.
- **Correct Nullifier Derivation:** It ensures the public nullifier is correctly computed from a secret seed. For government entities, this seed is derived from a `PoseidonHash` of the unique and stable `nicCert_domainName` and `nicCert_serialNumber`. This is vital to prevent the same department from claiming multiple Green Badges.

5.5 ZKP Circuit Design Logic for Private Company

This circuit enables an authorized representative of a private company to prove their social media handle’s association with their legally registered corporate identity, anchored by their Corporate Identification Number (CIN).

5.5.1 Objective (Company Verification)

To prove in zero-knowledge:

- The social media handle belongs to a genuine company registered with the Ministry of Corporate Affairs (MCA).
- The claim is attested to by an authorized signatory of the company using a valid DSC.
- The verification is unique to the company’s CIN.

5.5.2 Key Private Inputs (Processed Client-Side)

- The company’s official `coi_companyName` and its unique 21-digit `coi_cin` (Corporate Identification Number) as sourced from its Certificate of Incorporation (COI).
- Components from the authorized signatory’s DSC, including their signature over an attestation message that links the company name, CIN, and the claimed social media handle.

5.5.3 Key Public Inputs/Outputs (for ZKP Verification)

- The `claimed_socialMediaHandle` and the `claimed_companyCIN`.
- A cryptographic commitment to the relevant COI fields.
- The mandatory nullifier unique to the company.

5.5.4 Core Assertions in Circuit (High-Level Logic)

- **Corporate Identity Validation:** The circuit asserts that the private `coi_cin` matches the publicly claimed `claimed_companyCIN` and corresponds to the public commitment hash of the COI data.
- **DSC Attestation:** Similar to the government flow, the circuit verifies the signatory’s DSC signature over the attestation message and checks the validity of the issuing CA.
- **Correct Nullifier Derivation:** It enforces that the public nullifier is correctly derived from a secret seed, which for companies is computed from a `PoseidonHash` of the unique `coi_cin`. This ensures a company cannot acquire multiple Green Badges for its single legal identity.

5.6 Nullifier Scheme

The nullifier scheme is vital for preventing a single authentic entity from obtaining multiple Green Badges for the same core identity, thus mitigating Sybil attacks on this high-assurance tier.

Purpose: To create a unique, publicly revealable "spent token" for each verified identity.

Derivation: As described in the circuit logic, the nullifier is derived within the ZKP by first computing a *secret_seed* from unique and stable private identifiers obtained from the verified authoritative document, and then hashing this seed with an entity-type-specific salt using Poseidon.

- **Individual:** Seed from PoseidonHash (*Name, DoB, Gender, Last4Digits from verifiedAadhaarXML*).
- **Government:** Seed from PoseidonHash (*NormalizedDomainName, NICCertSerialNumber*).
- **Company:** Seed from PoseidonHash (Company CIN).

The ZKP guarantees that the submitted nullifier is correctly derived from the same credentials being attested to in the proof.

6 Intuition Security Analysis

The proposed system's security relies on the cryptographic properties of ZKPs and the designed protocol flow.

6.1 Resilience Against Key Attacks

- **Impersonation:** Mitigated by requiring ZKPs based on authentic, cryptographically signed Indian identity documents (Aadhaar, NIC/MCA certs with DSCs). The ZKP proves control/possession of these credentials and the validity of claims derived from them.
- **Fraudulent Proofs:** The computational soundness of Groth16 (assuming a secure trusted setup) makes it infeasible to generate valid proofs for false statements.
- **Privacy Breaches of Inputs:** The zero-knowledge property ensures the platform does

not see sensitive private data from user documents; only the proof, public commitments, and nullifier are shared.

- **Nullifier Scheme:** The use of unique nullifiers (derived from stable, verified data like CIN or Aadhaar demographics, and cryptographically hashed) effectively prevents double registration for the same identity, with the ZKP ensuring correct derivation.
- **Trusted Setup Implication:** The security of Groth16 against fake proofs is contingent on the integrity of the per-circuit trusted setup. Production systems require robust, transparent Multi-Party Computation (MPC) for this.
- **External Dependencies:** The system's overall trust relies on the security of the underlying Indian identity document issuance processes (UIDAI, NIC, MCA, CCA).

7 Privacy Preservation

A core design goal is maximizing user privacy during verification. This is achieved through:

- **Zero-Knowledge Proofs:** By definition, ZKPs allow proof of a statement's validity without revealing the private data (witness) that makes it true. The platform does not see the raw contents of the Aadhaar eKYC XML, NIC, or MCA documents.
- **Client-Side Processing:** All parsing of sensitive documents, extraction of private information, and witness generation occurs locally on the user's device. Only the ZKP, pre-defined public inputs/commitments, and the nullifier are transmitted to the platform.
- **Data Minimization via Commitments:** The platform can record public cryptographic commitments (hashes) to verified data (e.g., `commitment_AadhaarDemographics`) instead of storing the actual PII, if its policy chooses further minimization post-verification.
- **Nullifier Privacy:** The nullifier, while recorded by the platform, is a hash of a secret seed derived from private user data and does not directly reveal that underlying data.

7.1 Scalability

The system is designed with scalability in mind:

- **Succinct Proofs:** Groth16 proofs are small (~ 1 KB) and constant-sized, minimizing network bandwidth and storage requirements, which is highly beneficial for large user bases.
- **Fast Verification:** Proof verification is computationally inexpensive (milliseconds), allowing the platform to handle a high volume of verification requests by horizontally scaling stateless verification servers.
- **User-Side Proof Generation:** This is the most computationally intensive step for the user. While PoC times with `snarkjs` were in seconds, these are for a one-time verification event. Optimizations through native provers and further circuit refinement are expected to significantly reduce these times in a production environment.
- **Nullifier Database:** Requires a high-throughput database for uniqueness checks at scale, which is a standard, solvable engineering challenge for large platforms.

Overall, the ZKP-based approach is conceptually well-suited for large-scale deployment, with the main optimization focus being on the client-side prover efficiency.

8 Conclusion and Future Work

8.1 Conclusion

Social media platforms have become indispensable for communication and information sharing, yet their utility is increasingly compromised by an erosion of trust stemming from ambiguous and easily manipulated account verification systems. The shift towards paid verification badges on platforms like X (formerly Twitter) has exacerbated user confusion, making it difficult to discern authentic accounts from those merely subscribing to premium features, thereby elevating risks of impersonation and the spread of misinformation. This research addressed the urgent need for a clear, scalable, trustworthy, and privacy-preserving mechanism for verifying authentic identities online.

This project proposed and detailed a novel, differentiated two-tier verification framework featuring a basic subscription-based "Blue Badge" and a high-assurance "Green Badge" signifying cryptographically

verified authenticity using Zero-Knowledge Proofs (ZKPs), specifically Groth16. As a proof-of-concept, ZKP circuits were designed in Circom to leverage the Indian digital identity ecosystem, demonstrating how government organizations (via NIC certificates and DSCs), private companies (via MCA COIs and DSCs), and recognized individuals (via UIDAI Aadhaar eKYC XML) can prove their authenticity without revealing sensitive underlying data to the social media platform. The developed system, including a robust nullifier scheme to prevent double-registration and a detailed cryptographic protocol, offers a technically grounded pathway to enhance user trust, improve information integrity, and foster a more authentic online environment by providing unambiguous, verifiable signals of identity. The PoC implementation successfully demonstrated the feasibility of this approach, with preliminary performance indicating acceptable user-side proof generation times (optimizable with native provers) and highly efficient platform-side verification.

8.2 Future Work

This research successfully establishes a foundational framework and a strong proof-of-concept for ZKP-based social media verification. The findings and the inherent scope of this project open several impactful avenues for future work aimed at evolving this system into a globally applicable and production-ready solution.

- **From Proof-of-Concept to Production-Ready Implementation:** The current PoC successfully demonstrated feasibility using specific Indian identity documents and a `snarkjs`-based prover. The immediate next step is to move beyond this scope by developing high-performance native provers (e.g., in Rust or C++) for various client platforms (web, mobile) to significantly reduce proof generation times. This also involves creating robust client applications capable of parsing all variations of the target identity documents, and moving from a simulated trusted setup to a secure, transparent Multi-Party Computation (MPC) for key generation.
- **Generalization and Global Interoperability:** The practical implementation was intentionally tailored to the powerful Indian digital identity ecosystem. Future work should focus on extending the framework's global applicability by designing and implementing ZKP circuits

for authoritative identity documents from other countries and regions. This involves developing a modular architecture for easier integration and investigating alignment with emerging decentralized identity standards like W3C Verifiable Credentials (VCs) and Decentralized Identifiers (DIDs) to foster broader interoperability.

- **Comprehensive Human-Computer Interaction (HCI) Studies:** While this report proposes a clear UX distinction between the Blue and Green badges, large-scale empirical user testing was beyond the scope of this project. A crucial direction for future work is to conduct comprehensive usability studies to evaluate the real-world clarity and effectiveness of the differentiated badge system. Such studies should assess the Green Badge application process across diverse demographics and measure its tangible impact on user trust and information discernment.
- **Advanced Governance and Revocation Mechanisms:** This research conceptually discussed the need for advanced features but did not prototype them. Future work must address the design of efficient, privacy-preserving ZKP-based revocation schemes for Green Badges, allowing credentials to be invalidated without compromising user privacy. Furthermore, this involves proposing robust governance models for managing the ZKP circuits, trusted setups (potentially exploring universal schemes like PLONK to ease the operational burden), and the overall integrity of the Green Badge ecosystem.

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